Md. Fazle Rabbi Rithik

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LinkedIn GitHub Portfolio Codeforces

SUMMARY

Fresh Computer Science graduate from RUET and competitive programmer (Codeforces Specialist, max rating **1422**; **1300+** problems solved) with strong foundations in data structures, algorithms, OOP, and relational databases. Built database-backed web applications and REST APIs using Node.js, FastAPI, PostgreSQL, and MongoDB, including multi-role business workflows such as book borrowing and dataset review, with Git-based development and automated testing. Enjoys breaking problems down logically, understanding how real systems work, and picking up new stacks quickly.

EDUCATION

Rajshahi University of Engineering & Technology (RUET) March 2022 – August 2026
B.Sc. in Computer Science & Engineering **CGPA: 3.48 / 4.00**

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

- **Codeforces – Specialist (max rating 1422):** Solved **1300+** problems across Codeforces, LeetCode, CSES, and LightOJ.
- **ICPC Asia Dhaka Regional:** Ranked **73rd** with team `Ruet_Potato_Coders`. **2025**
- **DUET IUPC:** Ranked **73rd** with team `RUET_BRUTEFORCE`. **2025**
- **Problem Setter, CodeSmash:** Designed problems, statements, and test cases for an ICPC-format RUET fresher contest. **2022**
- **RUET Intra-University Contests:** Ranked **6th** in CODESPARK 2022 (team) and **6th** in CODESMASH 2021 (individual). **2021–2022**

TECHNICAL SKILLS

Programming Languages:	Python, C++, JavaScript, C
Backend & APIs:	Node.js, Express.js, FastAPI, REST APIs & HTTP, SQLAlchemy, Mongoose
Databases:	PostgreSQL, MongoDB, SQL (Joins, Subqueries, Aggregation, Constraints), Database Design & Data Modeling
Fundamentals:	Data Structures & Algorithms, OOP, Computer Networks
Tools & Practices:	Git, GitHub, Docker, Docker Compose, Linux, Automated Testing (Pytest)

PROJECTS

Annotation Wizard – Automated Dataset Processing Platform | *Python, FastAPI, PostgreSQL, ONNX Runtime, Docker* Live Demo | GitHub

- Developed an automated data-processing platform that accepts image datasets, performs ONNX-based model inference, and generates structured annotations for export.
- Designed PostgreSQL database models for dataset jobs and review workflows, allowing users to inspect and modify records, edit annotations, and export processed results.
- Built REST APIs with API-key authentication and PostgreSQL persistence; configured and deployed application services using Docker Compose.
- Implemented automated tests using Pytest to verify application functionality and API behavior.

Library Management System | *Node.js, Express.js, MongoDB, Mongoose, JavaScript* GitHub

- Developed a full-stack application that turns real library operations into software, with separate admin and student workflows for inventory, browsing, borrowing, returns, and borrowing-history tracking.
- Designed MongoDB schemas for users, books, and borrowing records, and implemented REST APIs to manage application data and business workflows.
- Implemented user registration and login with bcrypt-based password hashing.

UNDERGRADUATE THESIS

Class-Aware Data Augmentation for Vision-Based Road Damage Detection: A Systematic Ablation Study

- Conducted 8+ controlled YOLOv8 experiments on augmentation strategies; analysed precision, recall, and mAP@0.5 across four damage classes to draw evidence-based conclusions.